Raw JSON Results — Stage 1

FIT4701 Semester 1 2026

research_results_12.json


```
{
  "two_sum__openai__basic": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 104,
      "output_tokens": 185,
      "total_tokens": 289,
      "elapsed_s": 2.648,
      "tokens_per_sec": 69.9,
      "cost": 0.0009105,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 104,
      "output_tokens": 220,
      "total_tokens": 324,
      "elapsed_s": 2.483,
      "tokens_per_sec": 88.6,
      "cost": 0.001068,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 104,
      "output_tokens": 206,
      "total_tokens": 310,
      "elapsed_s": 1.669,
      "tokens_per_sec": 123.4,
      "cost": 0.001005,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 104,
      "output_tokens": 185,
      "total_tokens": 289,
      "elapsed_s": 1.9,
      "tokens_per_sec": 97.3,
      "cost": 0.0009105,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 104,
      "output_tokens": 182,
      "total_tokens": 286,
      "elapsed_s": 3.64,
      "tokens_per_sec": 50.0,
      "cost": 0.000897,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "two_sum__openai__efficiency": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 137,
      "output_tokens": 310,
      "total_tokens": 447,
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      "elapsed_s": 2.889,
      "tokens_per_sec": 107.3,
      "cost": 0.00149775,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 137,
      "output_tokens": 276,
      "total_tokens": 413,
      "elapsed_s": 2.463,
      "tokens_per_sec": 112.1,
      "cost": 0.00134475,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 137,
      "output_tokens": 287,
      "total_tokens": 424,
      "elapsed_s": 2.533,
      "tokens_per_sec": 113.3,
      "cost": 0.00139425,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 137,
      "output_tokens": 303,
      "total_tokens": 440,
      "elapsed_s": 3.012,
      "tokens_per_sec": 100.6,
      "cost": 0.00146625,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 137,
      "output_tokens": 311,
      "total_tokens": 448,
      "elapsed_s": 2.97,
      "tokens_per_sec": 104.7,
      "cost": 0.00150225,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "two_sum__openai__two_step": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 336,
      "output_tokens": 384,
      "total_tokens": 720,
      "elapsed_s": 3.537,
      "tokens_per_sec": 108.6,
      "cost": 0.00198,
      "step1_total_tok": 284,
      "step2_total_tok": 436,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 326,
      "output_tokens": 376,
      "total_tokens": 702,
      "elapsed_s": 4.909,
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      "tokens_per_sec": 76.6,
      "cost": 0.0019365,
      "step1_total_tok": 274,
      "step2_total_tok": 428,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 340,
      "output_tokens": 380,
      "total_tokens": 720,
      "elapsed_s": 4.142,
      "tokens_per_sec": 91.7,
      "cost": 0.001965,
      "step1_total_tok": 288,
      "step2_total_tok": 432,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 326,
      "output_tokens": 371,
      "total_tokens": 697,
      "elapsed_s": 12.584,
      "tokens_per_sec": 29.5,
      "cost": 0.001914,
      "step1_total_tok": 274,
      "step2_total_tok": 423,
      "tests_passed": 3,
      "tests_total": 3
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      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 343,
      "output_tokens": 425,
      "total_tokens": 768,
      "elapsed_s": 3.749,
      "tokens_per_sec": 113.4,
      "cost": 0.00216975,
      "step1_total_tok": 291,
      "step2_total_tok": 477,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "two_sum__anthropic__basic": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 106,
      "output_tokens": 556,
      "total_tokens": 662,
      "elapsed_s": 4.094,
      "tokens_per_sec": 135.8,
      "cost": 0.002886,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 106,
      "output_tokens": 556,
      "total_tokens": 662,
      "elapsed_s": 4.065,
      "tokens_per_sec": 136.8,
      "cost": 0.002886,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 106,
      "output_tokens": 503,
      "total_tokens": 609,
      "elapsed_s": 3.811,
      "tokens_per_sec": 132.0,
      "cost": 0.002621,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 106,
      "output_tokens": 505,
      "total_tokens": 611,
      "elapsed_s": 3.933,
      "tokens_per_sec": 128.4,
      "cost": 0.002631,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 106,
      "output_tokens": 503,
      "total_tokens": 609,
      "elapsed_s": 5.926,
      "tokens_per_sec": 84.9,
      "cost": 0.002621,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "two_sum__anthropic__efficiency": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 143,
      "output_tokens": 682,
      "total_tokens": 825,
      "elapsed_s": 5.806,
      "tokens_per_sec": 117.5,
      "cost": 0.003553,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 143,
      "output_tokens": 700,
      "total_tokens": 843,
      "elapsed_s": 5.555,
      "tokens_per_sec": 126.0,
      "cost": 0.003643,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 143,
      "output_tokens": 622,
      "total_tokens": 765,
      "elapsed_s": 5.404,
      "tokens_per_sec": 115.1,
      "cost": 0.003253,
      "tests_passed": 3,
      "tests_total": 3
    },
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      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 143,
      "output_tokens": 695,
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      "total_tokens": 838,
      "elapsed_s": 6.143,
      "tokens_per_sec": 113.1,
      "cost": 0.003618,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 143,
      "output_tokens": 847,
      "total_tokens": 990,
      "elapsed_s": 7.1,
      "tokens_per_sec": 119.3,
      "cost": 0.004378,
      "tests_passed": 3,
      "tests_total": 3
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  ],
  "two_sum__anthropic__two_step": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 663,
      "output_tokens": 863,
      "total_tokens": 1526,
      "elapsed_s": 7.998,
      "tokens_per_sec": 107.9,
      "cost": 0.004978,
      "step1_total_tok": 609,
      "step2_total_tok": 917,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 665,
      "output_tokens": 881,
      "total_tokens": 1546,
      "elapsed_s": 7.415,
      "tokens_per_sec": 118.8,
      "cost": 0.00507,
      "step1_total_tok": 611,
      "step2_total_tok": 935,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 663,
      "output_tokens": 866,
      "total_tokens": 1529,
      "elapsed_s": 7.659,
      "tokens_per_sec": 113.1,
      "cost": 0.004993,
      "step1_total_tok": 609,
      "step2_total_tok": 920,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 663,
      "output_tokens": 882,
      "total_tokens": 1545,
      "elapsed_s": 7.555,
      "tokens_per_sec": 116.7,
      "cost": 0.005073,
      "step1_total_tok": 609,
      "step2_total_tok": 936,
      "tests_passed": 3,
      "tests_total": 3
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    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 663,
      "output_tokens": 844,
      "total_tokens": 1507,
      "elapsed_s": 7.357,
      "tokens_per_sec": 114.7,
      "cost": 0.004883,
      "step1_total_tok": 609,
      "step2_total_tok": 898,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "max_subarray__openai__basic": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 87,
      "output_tokens": 242,
      "total_tokens": 329,
      "elapsed_s": 2.094,
      "tokens_per_sec": 115.6,
      "cost": 0.00115425,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 87,
      "output_tokens": 224,
      "total_tokens": 311,
      "elapsed_s": 2.229,
      "tokens_per_sec": 100.5,
      "cost": 0.00107325,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 87,
      "output_tokens": 240,
      "total_tokens": 327,
      "elapsed_s": 4.226,
      "tokens_per_sec": 56.8,
      "cost": 0.00114525,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 87,
      "output_tokens": 239,
      "total_tokens": 326,
      "elapsed_s": 1.984,
      "tokens_per_sec": 120.4,
      "cost": 0.00114075,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 87,
      "output_tokens": 258,
      "total_tokens": 345,
      "elapsed_s": 2.302,
      "tokens_per_sec": 112.1,
      "cost": 0.00122625,
      "tests_passed": 3,
      "tests_total": 3
    }
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],
"max_subarray__openai__efficiency": [
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 121,
    "output_tokens": 312,
    "total_tokens": 433,
    "elapsed_s": 3.01,
    "tokens_per_sec": 103.7,
    "cost": 0.00149475,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 121,
    "output_tokens": 386,
    "total_tokens": 507,
    "elapsed_s": 3.189,
    "tokens_per_sec": 121.1,
    "cost": 0.00182775,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 121,
    "output_tokens": 324,
    "total_tokens": 445,
    "elapsed_s": 5.147,
    "tokens_per_sec": 63.0,
    "cost": 0.00154875,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 121,
    "output_tokens": 311,
    "total_tokens": 432,
    "elapsed_s": 2.419,
    "tokens_per_sec": 128.6,
    "cost": 0.00149025,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 121,
    "output_tokens": 327,
    "total_tokens": 448,
    "elapsed_s": 2.692,
    "tokens_per_sec": 121.5,
    "cost": 0.00156225,
    "tests_passed": 3,
    "tests_total": 3
  }
],
"max_subarray__openai__two_step": [
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 377,
    "output_tokens": 445,
    "total_tokens": 822,
    "elapsed_s": 4.273,
    "tokens_per_sec": 104.1,
    "cost": 0.00228525,
    "step1_total_tok": 325,
    "step2_total_tok": 497,
    "tests_passed": 3,
    "tests_total": 3
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    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 347,
      "output_tokens": 395,
      "total_tokens": 742,
      "elapsed_s": 4.168,
      "tokens_per_sec": 94.8,
      "cost": 0.00203775,
      "step1_total_tok": 295,
      "step2_total_tok": 447,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 395,
      "output_tokens": 476,
      "total_tokens": 871,
      "elapsed_s": 3.922,
      "tokens_per_sec": 121.4,
      "cost": 0.00243825,
      "step1_total_tok": 344,
      "step2_total_tok": 527,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 361,
      "output_tokens": 419,
      "total_tokens": 780,
      "elapsed_s": 10.91,
      "tokens_per_sec": 38.4,
      "cost": 0.00215625,
      "step1_total_tok": 309,
      "step2_total_tok": 471,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 364,
      "output_tokens": 442,
      "total_tokens": 806,
      "elapsed_s": 5.265,
      "tokens_per_sec": 84.0,
      "cost": 0.002262,
      "step1_total_tok": 312,
      "step2_total_tok": 494,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "max_subarray__anthropic__basic": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 90,
      "output_tokens": 918,
      "total_tokens": 1008,
      "elapsed_s": 6.227,
      "tokens_per_sec": 147.4,
      "cost": 0.00468,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 90,
      "output_tokens": 834,
      "total_tokens": 924,
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      "elapsed_s": 5.136,
      "tokens_per_sec": 162.4,
      "cost": 0.00426,
      "tests_passed": 3,
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    },
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      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 90,
      "output_tokens": 839,
      "total_tokens": 929,
      "elapsed_s": 5.025,
      "tokens_per_sec": 167.0,
      "cost": 0.004285,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 90,
      "output_tokens": 834,
      "total_tokens": 924,
      "elapsed_s": 4.775,
      "tokens_per_sec": 174.7,
      "cost": 0.00426,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 90,
      "output_tokens": 919,
      "total_tokens": 1009,
      "elapsed_s": 5.354,
      "tokens_per_sec": 171.7,
      "cost": 0.004685,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "max_subarray__anthropic__efficiency": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 127,
      "output_tokens": 1086,
      "total_tokens": 1213,
      "elapsed_s": 6.433,
      "tokens_per_sec": 168.8,
      "cost": 0.005557,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 127,
      "output_tokens": 1078,
      "total_tokens": 1205,
      "elapsed_s": 6.951,
      "tokens_per_sec": 155.1,
      "cost": 0.005517,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 127,
      "output_tokens": 1069,
      "total_tokens": 1196,
      "elapsed_s": 6.737,
      "tokens_per_sec": 158.7,
      "cost": 0.005472,
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      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 127,
      "output_tokens": 770,
      "total_tokens": 897,
      "elapsed_s": 5.292,
      "tokens_per_sec": 145.5,
      "cost": 0.003977,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 127,
      "output_tokens": 1078,
      "total_tokens": 1205,
      "elapsed_s": 6.804,
      "tokens_per_sec": 158.4,
      "cost": 0.005517,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "max_subarray__anthropic__two_step": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 983,
      "output_tokens": 1162,
      "total_tokens": 2145,
      "elapsed_s": 8.3,
      "tokens_per_sec": 140.0,
      "cost": 0.0067929999999999996,
      "step1_total_tok": 929,
      "step2_total_tok": 1216,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1072,
      "output_tokens": 1311,
      "total_tokens": 2383,
      "elapsed_s": 9.141,
      "tokens_per_sec": 143.4,
      "cost": 0.007627,
      "step1_total_tok": 1018,
      "step2_total_tok": 1365,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1064,
      "output_tokens": 1301,
      "total_tokens": 2365,
      "elapsed_s": 9.394,
      "tokens_per_sec": 138.5,
      "cost": 0.007568999999999999,
      "step1_total_tok": 1010,
      "step2_total_tok": 1355,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1064,
      "output_tokens": 1446,
      "total_tokens": 2510,
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      "elapsed_s": 9.98,
      "tokens_per_sec": 144.9,
      "cost": 0.008294,
      "step1_total_tok": 1010,
      "step2_total_tok": 1500,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1116,
      "output_tokens": 1479,
      "total_tokens": 2595,
      "elapsed_s": 9.504,
      "tokens_per_sec": 155.6,
      "cost": 0.008511000000000001,
      "step1_total_tok": 1062,
      "step2_total_tok": 1533,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "sales_aggregation__openai__basic": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 354,
      "output_tokens": 452,
      "total_tokens": 806,
      "elapsed_s": 3.103,
      "tokens_per_sec": 145.7,
      "cost": 0.0022995,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 354,
      "output_tokens": 459,
      "total_tokens": 813,
      "elapsed_s": 3.271,
      "tokens_per_sec": 140.3,
      "cost": 0.002331,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 354,
      "output_tokens": 459,
      "total_tokens": 813,
      "elapsed_s": 4.672,
      "tokens_per_sec": 98.3,
      "cost": 0.002331,
      "tests_passed": 3,
      "tests_total": 3
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    {
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      "input_tokens": 354,
      "output_tokens": 453,
      "total_tokens": 807,
      "elapsed_s": 3.759,
      "tokens_per_sec": 120.5,
      "cost": 0.002304,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 354,
      "output_tokens": 481,
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    "total_tokens": 835,
    "elapsed_s": 3.041,
    "tokens_per_sec": 158.2,
    "cost": 0.00243,
    "tests_passed": 3,
    "tests_total": 3
  }
],
"sales_aggregation__openai__efficiency": [
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    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 387,
    "output_tokens": 622,
    "total_tokens": 1009,
    "elapsed_s": 4.415,
    "tokens_per_sec": 140.9,
    "cost": 0.00308925,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 387,
    "output_tokens": 579,
    "total_tokens": 966,
    "elapsed_s": 3.694,
    "tokens_per_sec": 156.8,
    "cost": 0.00289575,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 387,
    "output_tokens": 616,
    "total_tokens": 1003,
    "elapsed_s": 4.069,
    "tokens_per_sec": 151.4,
    "cost": 0.00306225,
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    "tests_total": 3
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    "input_tokens": 387,
    "output_tokens": 596,
    "total_tokens": 983,
    "elapsed_s": 3.832,
    "tokens_per_sec": 155.5,
    "cost": 0.00297225,
    "tests_passed": 3,
    "tests_total": 3
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    "model": "gpt-5.4-mini-2026-03-17",
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    "output_tokens": 582,
    "total_tokens": 969,
    "elapsed_s": 3.679,
    "tokens_per_sec": 158.2,
    "cost": 0.00290925,
    "tests_passed": 3,
    "tests_total": 3
  }
],
"sales_aggregation__openai__two_step": [
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    "input_tokens": 861,
    "output_tokens": 833,
    "total_tokens": 1694,
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      "elapsed_s": 5.635,
      "tokens_per_sec": 147.8,
      "cost": 0.004394250000000001,
      "step1_total_tok": 809,
      "step2_total_tok": 885,
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      "tests_total": 3
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      "input_tokens": 865,
      "output_tokens": 813,
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      "elapsed_s": 5.645,
      "tokens_per_sec": 144.0,
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      "step2_total_tok": 865,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 865,
      "output_tokens": 831,
      "total_tokens": 1696,
      "elapsed_s": 5.527,
      "tokens_per_sec": 150.4,
      "cost": 0.00438825,
      "step1_total_tok": 813,
      "step2_total_tok": 883,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 859,
      "output_tokens": 792,
      "total_tokens": 1651,
      "elapsed_s": 5.45,
      "tokens_per_sec": 145.3,
      "cost": 0.00420825,
      "step1_total_tok": 807,
      "step2_total_tok": 844,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 864,
      "output_tokens": 870,
      "total_tokens": 1734,
      "elapsed_s": 5.94,
      "tokens_per_sec": 146.5,
      "cost": 0.004563,
      "step1_total_tok": 812,
      "step2_total_tok": 922,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "sales_aggregation__anthropic__basic": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 394,
      "output_tokens": 986,
      "total_tokens": 1380,
      "elapsed_s": 5.923,
      "tokens_per_sec": 166.5,
      "cost": 0.005324,
      "tests_passed": 3,
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```
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 394,
      "output_tokens": 992,
      "total_tokens": 1386,
      "elapsed_s": 7.208,
      "tokens_per_sec": 137.6,
      "cost": 0.005354,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 394,
      "output_tokens": 962,
      "total_tokens": 1356,
      "elapsed_s": 5.319,
      "tokens_per_sec": 180.9,
      "cost": 0.005204,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 394,
      "output_tokens": 962,
      "total_tokens": 1356,
      "elapsed_s": 5.575,
      "tokens_per_sec": 172.5,
      "cost": 0.005204,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 394,
      "output_tokens": 986,
      "total_tokens": 1380,
      "elapsed_s": 5.982,
      "tokens_per_sec": 164.8,
      "cost": 0.005324,
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      "tests_total": 3
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      "input_tokens": 431,
      "output_tokens": 1190,
      "total_tokens": 1621,
      "elapsed_s": 9.63,
      "tokens_per_sec": 123.6,
      "cost": 0.006381,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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      "input_tokens": 431,
      "output_tokens": 1145,
      "total_tokens": 1576,
      "elapsed_s": 7.851,
      "tokens_per_sec": 145.8,
      "cost": 0.006156,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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```
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 431,
      "output_tokens": 1205,
      "total_tokens": 1636,
      "elapsed_s": 8.604,
      "tokens_per_sec": 140.0,
      "cost": 0.006456,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 431,
      "output_tokens": 1169,
      "total_tokens": 1600,
      "elapsed_s": 7.984,
      "tokens_per_sec": 146.4,
      "cost": 0.006276,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 431,
      "output_tokens": 1169,
      "total_tokens": 1600,
      "elapsed_s": 8.279,
      "tokens_per_sec": 141.2,
      "cost": 0.006276,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "sales_aggregation__anthropic__two_step": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1411,
      "output_tokens": 1886,
      "total_tokens": 3297,
      "elapsed_s": 10.594,
      "tokens_per_sec": 178.0,
      "cost": 0.010841,
      "step1_total_tok": 1357,
      "step2_total_tok": 1940,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1335,
      "output_tokens": 1806,
      "total_tokens": 3141,
      "elapsed_s": 11.425,
      "tokens_per_sec": 158.1,
      "cost": 0.010365,
      "step1_total_tok": 1281,
      "step2_total_tok": 1860,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1427,
      "output_tokens": 1850,
      "total_tokens": 3277,
      "elapsed_s": 10.481,
      "tokens_per_sec": 176.5,
      "cost": 0.010677,
      "step1_total_tok": 1373,
      "step2_total_tok": 1904,
      "tests_passed": 3,
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      "tests_total": 3
    },
    {
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      "input_tokens": 1410,
      "output_tokens": 1897,
      "total_tokens": 3307,
      "elapsed_s": 16.642,
      "tokens_per_sec": 114.0,
      "cost": 0.010895,
      "step1_total_tok": 1356,
      "step2_total_tok": 1951,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1434,
      "output_tokens": 1899,
      "total_tokens": 3333,
      "elapsed_s": 11.454,
      "tokens_per_sec": 165.8,
      "cost": 0.010929,
      "step1_total_tok": 1380,
      "step2_total_tok": 1953,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "data_cleaning__openai__basic": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 348,
      "output_tokens": 449,
      "total_tokens": 797,
      "elapsed_s": 2.945,
      "tokens_per_sec": 152.4,
      "cost": 0.0022815,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 348,
      "output_tokens": 447,
      "total_tokens": 795,
      "elapsed_s": 2.874,
      "tokens_per_sec": 155.5,
      "cost": 0.0022725,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 348,
      "output_tokens": 452,
      "total_tokens": 800,
      "elapsed_s": 4.428,
      "tokens_per_sec": 102.1,
      "cost": 0.002295,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 348,
      "output_tokens": 448,
      "total_tokens": 796,
      "elapsed_s": 2.909,
      "tokens_per_sec": 154.0,
      "cost": 0.002277,
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      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 348,
      "output_tokens": 441,
      "total_tokens": 789,
      "elapsed_s": 3.357,
      "tokens_per_sec": 131.4,
      "cost": 0.0022455,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "data_cleaning__openai__efficiency": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 381,
      "output_tokens": 631,
      "total_tokens": 1012,
      "elapsed_s": 4.397,
      "tokens_per_sec": 143.5,
      "cost": 0.00312525,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 381,
      "output_tokens": 612,
      "total_tokens": 993,
      "elapsed_s": 7.79,
      "tokens_per_sec": 78.6,
      "cost": 0.00303975,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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      "input_tokens": 381,
      "output_tokens": 656,
      "total_tokens": 1037,
      "elapsed_s": 3.867,
      "tokens_per_sec": 169.6,
      "cost": 0.00323775,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 381,
      "output_tokens": 613,
      "total_tokens": 994,
      "elapsed_s": 5.512,
      "tokens_per_sec": 111.2,
      "cost": 0.00304425,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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      "input_tokens": 381,
      "output_tokens": 607,
      "total_tokens": 988,
      "elapsed_s": 4.193,
      "tokens_per_sec": 144.8,
      "cost": 0.00301725,
      "tests_passed": 3,
      "tests_total": 3
    }
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],
"data_cleaning__openai__two_step": [
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 853,
    "output_tokens": 702,
    "total_tokens": 1555,
    "elapsed_s": 6.134,
    "tokens_per_sec": 114.4,
    "cost": 0.00379875,
    "step1_total_tok": 802,
    "step2_total_tok": 753,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 858,
    "output_tokens": 790,
    "total_tokens": 1648,
    "elapsed_s": 5.643,
    "tokens_per_sec": 140.0,
    "cost": 0.0041985,
    "step1_total_tok": 807,
    "step2_total_tok": 841,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 848,
    "output_tokens": 732,
    "total_tokens": 1580,
    "elapsed_s": 5.435,
    "tokens_per_sec": 134.7,
    "cost": 0.00393,
    "step1_total_tok": 797,
    "step2_total_tok": 783,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 839,
    "output_tokens": 792,
    "total_tokens": 1631,
    "elapsed_s": 5.791,
    "tokens_per_sec": 136.8,
    "cost": 0.00419325,
    "step1_total_tok": 788,
    "step2_total_tok": 843,
    "tests_passed": 3,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 828,
    "output_tokens": 755,
    "total_tokens": 1583,
    "elapsed_s": 5.236,
    "tokens_per_sec": 144.2,
    "cost": 0.0040185,
    "step1_total_tok": 776,
    "step2_total_tok": 807,
    "tests_passed": 3,
    "tests_total": 3
  }
],
"data_cleaning__anthropic__basic": [
  {
    "model": "claude-haiku-4-5-20251001",
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```
      "input_tokens": 372,
      "output_tokens": 933,
      "total_tokens": 1305,
      "elapsed_s": 5.793,
      "tokens_per_sec": 161.0,
      "cost": 0.005037,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 372,
      "output_tokens": 857,
      "total_tokens": 1229,
      "elapsed_s": 10.079,
      "tokens_per_sec": 85.0,
      "cost": 0.004657,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 372,
      "output_tokens": 884,
      "total_tokens": 1256,
      "elapsed_s": 5.498,
      "tokens_per_sec": 160.8,
      "cost": 0.004792,
      "tests_passed": 3,
      "tests_total": 3
    },
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      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 372,
      "output_tokens": 905,
      "total_tokens": 1277,
      "elapsed_s": 5.796,
      "tokens_per_sec": 156.1,
      "cost": 0.004897,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 372,
      "output_tokens": 915,
      "total_tokens": 1287,
      "elapsed_s": 5.578,
      "tokens_per_sec": 164.0,
      "cost": 0.004947,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "data_cleaning__anthropic__efficiency": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 409,
      "output_tokens": 1061,
      "total_tokens": 1470,
      "elapsed_s": 7.972,
      "tokens_per_sec": 133.1,
      "cost": 0.005714,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 409,
      "output_tokens": 1063,
      "total_tokens": 1472,
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      "elapsed_s": 8.057,
      "tokens_per_sec": 131.9,
      "cost": 0.005724,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
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      "input_tokens": 409,
      "output_tokens": 1040,
      "total_tokens": 1449,
      "elapsed_s": 7.498,
      "tokens_per_sec": 138.7,
      "cost": 0.005609,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 409,
      "output_tokens": 950,
      "total_tokens": 1359,
      "elapsed_s": 6.569,
      "tokens_per_sec": 144.6,
      "cost": 0.005159,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 409,
      "output_tokens": 1046,
      "total_tokens": 1455,
      "elapsed_s": 7.548,
      "tokens_per_sec": 138.6,
      "cost": 0.005639,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
  "data_cleaning__anthropic__two_step": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1314,
      "output_tokens": 1702,
      "total_tokens": 3016,
      "elapsed_s": 11.813,
      "tokens_per_sec": 144.1,
      "cost": 0.009824,
      "step1_total_tok": 1260,
      "step2_total_tok": 1756,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1359,
      "output_tokens": 1770,
      "total_tokens": 3129,
      "elapsed_s": 11.53,
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      "cost": 0.010209,
      "step1_total_tok": 1305,
      "step2_total_tok": 1824,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1283,
      "output_tokens": 1423,
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      "total_tokens": 2706,
      "elapsed_s": 10.912,
      "tokens_per_sec": 130.4,
      "cost": 0.008398,
      "step1_total_tok": 1229,
      "step2_total_tok": 1477,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1378,
      "output_tokens": 1882,
      "total_tokens": 3260,
      "elapsed_s": 13.326,
      "tokens_per_sec": 141.2,
      "cost": 0.010787999999999999,
      "step1_total_tok": 1324,
      "step2_total_tok": 1936,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1342,
      "output_tokens": 1765,
      "total_tokens": 3107,
      "elapsed_s": 11.461,
      "tokens_per_sec": 154.0,
      "cost": 0.010166999999999999,
      "step1_total_tok": 1288,
      "step2_total_tok": 1819,
      "tests_passed": 3,
      "tests_total": 3
    }
  ]
}
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research_results_3.json


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{
  "weather_fetch__openai__basic": [
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 205,
      "output_tokens": 314,
      "total_tokens": 519,
      "elapsed_s": 3.387,
      "tokens_per_sec": 92.7,
      "cost": 0.00156675,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 205,
      "output_tokens": 298,
      "total_tokens": 503,
      "elapsed_s": 2.409,
      "tokens_per_sec": 123.7,
      "cost": 0.00149475,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 205,
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      "elapsed_s": 3.038,
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      "tokens_per_sec": 103.7,
      "cost": 0.00157125,
      "tests_passed": 2,
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      "input_tokens": 205,
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      "tests_passed": 2,
      "tests_total": 3
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      "input_tokens": 205,
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      "elapsed_s": 2.802,
      "tokens_per_sec": 118.9,
      "cost": 0.00165225,
      "tests_passed": 2,
      "tests_total": 3
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  "weather_fetch__openai__efficiency": [
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      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 247,
      "output_tokens": 372,
      "total_tokens": 619,
      "elapsed_s": 4.436,
      "tokens_per_sec": 83.9,
      "cost": 0.00185925,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 247,
      "output_tokens": 383,
      "total_tokens": 630,
      "elapsed_s": 3.313,
      "tokens_per_sec": 115.6,
      "cost": 0.00190875,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 247,
      "output_tokens": 433,
      "total_tokens": 680,
      "elapsed_s": 3.982,
      "tokens_per_sec": 108.7,
      "cost": 0.00213375,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 247,
      "output_tokens": 402,
      "total_tokens": 649,
      "elapsed_s": 5.159,
      "tokens_per_sec": 77.9,
      "cost": 0.00199425,
      "tests_passed": 2,
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      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 247,
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      "tokens_per_sec": 97.9,
      "cost": 0.00181875,
      "tests_passed": 2,
      "tests_total": 3
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      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 588,
      "output_tokens": 1098,
      "total_tokens": 1686,
      "elapsed_s": 7.547,
      "tokens_per_sec": 145.5,
      "cost": 0.005382,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 578,
      "output_tokens": 1041,
      "total_tokens": 1619,
      "elapsed_s": 7.844,
      "tokens_per_sec": 132.7,
      "cost": 0.005118,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 590,
      "output_tokens": 943,
      "total_tokens": 1533,
      "elapsed_s": 6.836,
      "tokens_per_sec": 137.9,
      "cost": 0.004686,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 561,
      "output_tokens": 1068,
      "total_tokens": 1629,
      "elapsed_s": 7.804,
      "tokens_per_sec": 136.9,
      "cost": 0.00522675,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 585,
      "output_tokens": 905,
      "total_tokens": 1490,
      "elapsed_s": 7.486,
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      "cost": 0.00451125,
      "tests_passed": 3,
      "tests_total": 3
    }
  ],
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"weather_fetch__anthropic__basic": [
  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 228,
    "output_tokens": 1173,
    "total_tokens": 1401,
    "elapsed_s": 7.356,
    "tokens_per_sec": 159.5,
    "cost": 0.006093,
    "tests_passed": 2,
    "tests_total": 3
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  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 228,
    "output_tokens": 1084,
    "total_tokens": 1312,
    "elapsed_s": 6.78,
    "tokens_per_sec": 159.9,
    "cost": 0.005648,
    "tests_passed": 2,
    "tests_total": 3
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  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 228,
    "output_tokens": 1106,
    "total_tokens": 1334,
    "elapsed_s": 6.939,
    "tokens_per_sec": 159.4,
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    "tests_passed": 2,
    "tests_total": 3
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    "input_tokens": 228,
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    "total_tokens": 1280,
    "elapsed_s": 6.758,
    "tokens_per_sec": 155.7,
    "cost": 0.005488,
    "tests_passed": 2,
    "tests_total": 3
  },
  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 228,
    "output_tokens": 1052,
    "total_tokens": 1280,
    "elapsed_s": 6.828,
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    "cost": 0.005488,
    "tests_passed": 2,
    "tests_total": 3
  }
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"weather_fetch__anthropic__efficiency": [
  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 273,
    "output_tokens": 1195,
    "total_tokens": 1468,
    "elapsed_s": 8.802,
    "tokens_per_sec": 135.8,
    "cost": 0.006248,
    "tests_passed": 2,
    "tests_total": 3
  },
  {
    "model": "claude-haiku-4-5-20251001",
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      "input_tokens": 273,
      "output_tokens": 1152,
      "total_tokens": 1425,
      "elapsed_s": 8.059,
      "tokens_per_sec": 142.9,
      "cost": 0.006033,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 273,
      "output_tokens": 1146,
      "total_tokens": 1419,
      "elapsed_s": 8.012,
      "tokens_per_sec": 143.0,
      "cost": 0.006003,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 273,
      "output_tokens": 1107,
      "total_tokens": 1380,
      "elapsed_s": 8.126,
      "tokens_per_sec": 136.2,
      "cost": 0.005808,
      "tests_passed": 2,
      "tests_total": 3
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    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 273,
      "output_tokens": 1111,
      "total_tokens": 1384,
      "elapsed_s": 7.669,
      "tokens_per_sec": 144.9,
      "cost": 0.005828,
      "tests_passed": 2,
      "tests_total": 3
    }
  ],
  "weather_fetch__anthropic__two_step": [
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1490,
      "output_tokens": 2844,
      "total_tokens": 4334,
      "elapsed_s": 17.711,
      "tokens_per_sec": 160.6,
      "cost": 0.015710000000000002,
      "tests_passed": 2,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
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      "output_tokens": 2538,
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      "elapsed_s": 16.065,
      "tokens_per_sec": 158.0,
      "cost": 0.014163000000000002,
      "tests_passed": 2,
      "tests_total": 3
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      "input_tokens": 1473,
      "output_tokens": 2524,
      "total_tokens": 3997,
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      "elapsed_s": 16.068,
      "tokens_per_sec": 157.1,
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    {
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      "input_tokens": 1395,
      "output_tokens": 2722,
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      "elapsed_s": 17.759,
      "tokens_per_sec": 153.3,
      "cost": 0.015005,
      "tests_passed": 2,
      "tests_total": 3
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    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1387,
      "output_tokens": 2561,
      "total_tokens": 3948,
      "elapsed_s": 18.057,
      "tokens_per_sec": 141.8,
      "cost": 0.014192,
      "tests_passed": 2,
      "tests_total": 3
    }
  ],
  "weather_compare__openai__basic": [
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      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 223,
      "output_tokens": 367,
      "total_tokens": 590,
      "elapsed_s": 2.741,
      "tokens_per_sec": 133.9,
      "cost": 0.00181875,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 223,
      "output_tokens": 392,
      "total_tokens": 615,
      "elapsed_s": 3.261,
      "tokens_per_sec": 120.2,
      "cost": 0.00193125,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 223,
      "output_tokens": 376,
      "total_tokens": 599,
      "elapsed_s": 3.171,
      "tokens_per_sec": 118.6,
      "cost": 0.00185925,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 223,
      "output_tokens": 357,
      "total_tokens": 580,
      "elapsed_s": 2.643,
      "tokens_per_sec": 135.1,
      "cost": 0.00177375,
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      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 223,
      "output_tokens": 368,
      "total_tokens": 591,
      "elapsed_s": 2.536,
      "tokens_per_sec": 145.1,
      "cost": 0.00182325,
      "tests_passed": 3,
      "tests_total": 3
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  ],
  "weather_compare__openai__efficiency": [
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      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 265,
      "output_tokens": 523,
      "total_tokens": 788,
      "elapsed_s": 3.943,
      "tokens_per_sec": 132.6,
      "cost": 0.00255225,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 265,
      "output_tokens": 515,
      "total_tokens": 780,
      "elapsed_s": 3.343,
      "tokens_per_sec": 154.1,
      "cost": 0.00251625,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 265,
      "output_tokens": 501,
      "total_tokens": 766,
      "elapsed_s": 4.089,
      "tokens_per_sec": 122.5,
      "cost": 0.00245325,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 265,
      "output_tokens": 541,
      "total_tokens": 806,
      "elapsed_s": 3.896,
      "tokens_per_sec": 138.9,
      "cost": 0.00263325,
      "tests_passed": 3,
      "tests_total": 3
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    {
      "model": "gpt-5.4-mini-2026-03-17",
      "input_tokens": 265,
      "output_tokens": 543,
      "total_tokens": 808,
      "elapsed_s": 3.916,
      "tokens_per_sec": 138.7,
      "cost": 0.00264225,
      "tests_passed": 3,
      "tests_total": 3
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],
"weather_compare__openai__two_step": [
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 641,
    "output_tokens": 916,
    "total_tokens": 1557,
    "elapsed_s": 6.588,
    "tokens_per_sec": 139.0,
    "cost": 0.00460275,
    "tests_passed": 1,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 663,
    "output_tokens": 928,
    "total_tokens": 1591,
    "elapsed_s": 6.741,
    "tokens_per_sec": 137.7,
    "cost": 0.00467325,
    "tests_passed": 1,
    "tests_total": 3
  },
  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 656,
    "output_tokens": 969,
    "total_tokens": 1625,
    "elapsed_s": 11.986,
    "tokens_per_sec": 80.8,
    "cost": 0.0048525,
    "tests_passed": 0,
    "tests_total": 3
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  {
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    "input_tokens": 656,
    "output_tokens": 1027,
    "total_tokens": 1683,
    "elapsed_s": 12.04,
    "tokens_per_sec": 85.3,
    "cost": 0.0051135,
    "tests_passed": 0,
    "tests_total": 3
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  {
    "model": "gpt-5.4-mini-2026-03-17",
    "input_tokens": 639,
    "output_tokens": 911,
    "total_tokens": 1550,
    "elapsed_s": 6.252,
    "tokens_per_sec": 145.7,
    "cost": 0.0045787499999999995,
    "tests_passed": 1,
    "tests_total": 3
  }
],
"weather_compare__anthropic__basic": [
  {
    "model": "claude-haiku-4-5-20251001",
    "input_tokens": 250,
    "output_tokens": 1157,
    "total_tokens": 1407,
    "elapsed_s": 6.394,
    "tokens_per_sec": 181.0,
    "cost": 0.006035,
    "tests_passed": 1,
    "tests_total": 3
  },
  {
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      "model": "claude-haiku-4-5-20251001",
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      "output_tokens": 1156,
      "total_tokens": 1406,
      "elapsed_s": 6.824,
      "tokens_per_sec": 169.4,
      "cost": 0.00603,
      "tests_passed": 1,
      "tests_total": 3
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      "output_tokens": 1148,
      "total_tokens": 1398,
      "elapsed_s": 6.863,
      "tokens_per_sec": 167.3,
      "cost": 0.00599,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 250,
      "output_tokens": 1149,
      "total_tokens": 1399,
      "elapsed_s": 6.315,
      "tokens_per_sec": 181.9,
      "cost": 0.005995,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 250,
      "output_tokens": 1137,
      "total_tokens": 1387,
      "elapsed_s": 6.443,
      "tokens_per_sec": 176.5,
      "cost": 0.005935,
      "tests_passed": 1,
      "tests_total": 3
    }
  ],
  "weather_compare__anthropic__efficiency": [
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      "input_tokens": 295,
      "output_tokens": 1421,
      "total_tokens": 1716,
      "elapsed_s": 10.688,
      "tokens_per_sec": 133.0,
      "cost": 0.0074,
      "tests_passed": 3,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 295,
      "output_tokens": 1174,
      "total_tokens": 1469,
      "elapsed_s": 8.525,
      "tokens_per_sec": 137.7,
      "cost": 0.006165,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 295,
      "output_tokens": 1174,
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      "total_tokens": 1469,
      "elapsed_s": 9.535,
      "tokens_per_sec": 123.1,
      "cost": 0.006165,
      "tests_passed": 1,
      "tests_total": 3
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    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 295,
      "output_tokens": 1197,
      "total_tokens": 1492,
      "elapsed_s": 8.483,
      "tokens_per_sec": 141.1,
      "cost": 0.00628,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
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      "input_tokens": 295,
      "output_tokens": 1174,
      "total_tokens": 1469,
      "elapsed_s": 8.483,
      "tokens_per_sec": 138.4,
      "cost": 0.006165,
      "tests_passed": 1,
      "tests_total": 3
    }
  ],
  "weather_compare__anthropic__two_step": [
    {
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      "input_tokens": 1479,
      "output_tokens": 2749,
      "total_tokens": 4228,
      "elapsed_s": 15.935,
      "tokens_per_sec": 172.5,
      "cost": 0.015224,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1443,
      "output_tokens": 2823,
      "total_tokens": 4266,
      "elapsed_s": 19.5,
      "tokens_per_sec": 144.8,
      "cost": 0.015557999999999999,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1376,
      "output_tokens": 2550,
      "total_tokens": 3926,
      "elapsed_s": 15.315,
      "tokens_per_sec": 166.5,
      "cost": 0.014126,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1548,
      "output_tokens": 3042,
      "total_tokens": 4590,
      "elapsed_s": 18.384,
      "tokens_per_sec": 165.5,
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      "cost": 0.016758000000000002,
      "tests_passed": 1,
      "tests_total": 3
    },
    {
      "model": "claude-haiku-4-5-20251001",
      "input_tokens": 1480,
      "output_tokens": 2791,
      "total_tokens": 4271,
      "elapsed_s": 15.725,
      "tokens_per_sec": 177.5,
      "cost": 0.015435,
      "tests_passed": 1,
      "tests_total": 3
    }
  ]
}
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